

V02 ERGONOMIC WORKSTATION DESIGN | P

Intent : Reduce the risk of physical strain on the body through ergonomic design at workstations that supports neutral body positions for seated and standing work and provides opportunities to alternate between seated and standing positions.

Summary : This WELL feature requires projects to provide ergonomic workstation furnishings to accommodate all users, that allow for customized workstation fit and provide user orientation to workstations covering ergonomic workstation design and adjustability features.

Issue : In 2016, musculoskeletal disorders (MSDs) ranked among the top drivers of global disability.^{1,2} MSDs are one of the most commonly reported causes of lost or restricted work time and also contribute to absenteeism and low productivity.^{3,4} Risk factors in the workplace vary by the type of tasks being performed. In manual labor work environments, risk factors include heavy lifting, bending, reaching overhead and pushing or pulling heavy objects.⁴ In office settings, risk factors are no less prevalent and include workstation design that forces the body into awkward positions along with other occupational factors that expose the body to prolonged or repetitive tasks.⁵

Solutions : An ideal ergonomic work environment is conducive to the necessary breadth of tasks assigned to that space, while encouraging movement through a variety of positions throughout the day. Effective ergonomic interventions to accommodate all users include both design (e.g., adjustable furniture) and programmatic (e.g., education) approaches.^{6,7} Ergonomic design solutions facilitate customizability at workstations allowing users to better fit workstations to their needs. Preliminary studies have demonstrated an ROI for ergonomics interventions. One study found a return of \$10 USD for every \$1 USD invested.⁸ A second study examining outcomes across 250 case studies found generally positive results, including a reduction in the number (49.5% across 37 studies) and cost (64.8% across 22 studies) of work-related MSDs, and also noted that the payback period was generally less than one year.⁹

PART 1 SUPPORT VISUAL ERGONOMICS

For Office Spaces:

The project meets the following requirements:

- a. Workstations where desktop computers are used provide support for user adjustability (monitor height, viewing angle, horizontal distance) through one of the following:
 1. Monitors with built-in height and angle adjustment.^{10,11}
 2. Monitor stands or arms that allow height, angle and horizontal adjustment.^{10,11}
- b. Workstations where laptops are primarily used provide support for user adjustability through at least one of the following:
 1. An external keyboard, mouse and laptop stand such that the laptop screen can be positioned by the user (screen height, viewing angle, horizontal distance).¹¹
 2. An external monitor that meets requirement a.¹¹

PART 2 PROVIDE HEIGHT-ADJUSTABLE WORK SURFACES

For Office Spaces:

At least 25% of all workstations can be adjusted by the user for both seated and standing work, through one of the following:

- a. Manual or electric height-adjustable work surfaces that provide users with the ability to customize workstation height at both seated and standing positions.^{10,11}
- b. Supplemental solutions (e.g., stand) that allow all or part of the work surface, monitor and primary input devices (e.g., keyboard, mouse) to be raised or lowered to seated or standing heights.^{10,11}

PART 3 PROVIDE CHAIR ADJUSTABILITY

For Office Spaces:

All seating at workstations meets the following requirements:

- a. The seat height is adjustable.^{10,11}
- b. One of the following:¹⁰
 - i. The seat pan depth is adjustable.
 - ii. The seat pan depth is 43 cm or less.
- c. One of the following:^{10,11}
 - i. The backrest lumbar support is height adjustable.
 - ii. The backrest angle is adjustable.
 - iii. The armrest height and distance between armrests are adjustable.

PART 4 PROVIDE SUPPORT AT STANDING WORKSTATIONS

For All Spaces except Commercial Kitchen Spaces:

Option 1: Support for standing workers

All workstations in which users are regularly required to stand for 50% or more of their working hours (e.g., assembly line station, hotel check-in counter, supermarket check-out counter) incorporate at least two of the following:

- a. Anti-fatigue mats, impact reducing flooring or a similar strategy.¹²
- b. Recessed toe space of at least 10 cm depth and height.¹³
- c. A footrest or footrail.^{12,14}
- d. A sit-stand stool or high stool.^{12,14}

OR

Option 2: No standing workers

The project meets the following requirement:

- a. There are no workstations at which users are regularly required to stand for 50% or more of their working hours.

PART 5 PROVIDE WORKSTATION ORIENTATION

For All Spaces:

The following requirement is met:

- a. All eligible employees receive an orientation (e.g., in-person or virtual training, self-guided digital training) to workstations in the space covering, at minimum, the following:
 - 1. Adjustability features of all available workstation types (as applicable) and their benefits to users (e.g., customized fit for individual comfort).
 - 2. Instructions on how to make adjustments to achieve the intended benefits (e.g., customized fit for individual comfort).
 - 3. Orientation resources that can be used for future reference.

REFERENCES

1. Institute for Health Metrics and Evaluation (IHME). GBD Compare. 2015. <http://vizhub.healthdata.org/gbd-compare>.
2. Agarwal S, Steinmaus C, Harris-Adamson C. Sit-stand workstations and impact on low back discomfort: a systematic review and meta-analysis. *Ergonomics*. 2018. doi:10.1080/00140139.2017.1402960
3. Bevan S. Economic impact of musculoskeletal disorders (MSDs) on work in Europe. *Best Pract Res Clin Rheumatol*. 2015;29(3):356-373. doi:10.1016/j.berh.2015.08.002
4. Occupational Safety and Health Administration. *Ergonomics - Overview*. <https://www.osha.gov/SLTC/ergonomics/>. Accessed October 31, 2017.
5. Hedge A. Ergonomic Workplace Design for Health, Wellness, and Productivity. In: *Ergonomic Workplace Design for Health, Wellness, and Productivity*. ; 2016:1-443. doi:10.1201/9781315374000
6. American National Standards Institute (ANSI), Canadian Standards Association. CSA Z412-2017: Office Ergonomics An Application Standard For Workplace Ergonomics. Presented at the: 2017. https://webstore.ansi.org/Standards/CSA/CSAZ4122017?gclid=CjwKCAjw_qb3BRAVEiwAvwq6Vr6zDZ6zG1pch-2J-MkwBlZnFR5bVyD9Rfd9vQlg51GY8Ua47lU4BoCFelQAvD_BwE.
7. U.S. General Accounting Office. Worker Protection: Private Sector Ergonomics Programs Yield Positive Results. Washington, DC; 1997. <http://www.gao.gov/assets/230/224570.pdf>.
8. Heller-ono A. A Prospective Study of a Macroergonomics Process over Five Years Demonstrates Significant Prevention of Workers' Compensation Claims Resulting in Projected Savings. 2009:1-4. <http://worksiteinternational.com/pdf/ODAM-NES-2014-ARH-paper-Final-Submission.pdf>.
9. Goggins RW, Spielholz P, Nothstein GL. Estimating the effectiveness of ergonomics interventions through case studies: Implications for predictive cost-benefit analysis. *J Safety Res*. 2008;39(3):339-344. doi:10.1016/J.JSR.2007.12.006
10. Human Factors and Ergonomics Society. ANSI/HFES 100-2007 - Human Factors Engineering of Computer Workstations. Human Factors and Ergonomics Society; 2007. <https://www.hfes.org/Order/PublicationDetailsByProductId?ProductId=69>.
11. Business and Institutional Furniture Manufacturers Association. ANSI/BIFMA X10.1-2024 - Ergonomics Requirements for Furniture Designed for Computer Use; 2024. <https://www.madcad.com/store/subscription/BIFMA-X10.1-2024/>.
12. Ebben JM. Improved Ergonomics for Standing Work. *Occup Saf Heal*. 2003;72(4):72-76.
13. U.S. Occupational Safety and Health Administration. Guidelines for Retail Grocery Stores: Ergonomics for the Prevention of Musculoskeletal Disorders. 2004.

<https://www.osha.gov/ergonomics/guidelines/retailgrocery/retailgrocery.html#front>.

14. Canadian Centre for Occupational Health and Safety. Working in a Standing Position - Basic Information.
https://www.ccohs.ca/oshanswers/ergonomics/standing/standing_basic.html. Accessed May 15, 2018.